

Profile

Hands-on machine learning engineer with end-to-end experience in robotics and 3D vision. **PhD research** centered on multi-view 3D reconstruction, sensor fusion, and deep learning for geometric perception, specialized in **predictive modeling and transformer model for data-driven decision making**. **Strong software engineering skills** across algorithms, application development, and data workflows; write clean, maintainable code and deliver reliably with testing and automation. **Full-stack developing experience** to move ideas from prototype to production. **Proactive engineer with an analytical mindset** who turns ambiguous problems into clear metrics, dashboards, and experiments, and communicates insights to guide data-driven decisions. **Passionate about using data and machine learning to deliver measurable improvements in business performance and user experience.**

Technical Skills	Soft Skills
• Languages: Python, JavaScript, C++, C#, Java • GenAI: LLMs, Retrieval-Augmented Generation (RAG) • Machine Learning: PyTorch, TensorFlow, Scikit-learn • Data Platform: PostgreSQL, VectorDB, Databricks, Snowflake, Spark • Experimentation: A/B Test, Power Analysis, Statistic Tests • Full-stack Development: React, Node.js, RESTful APIs, GraphQL • DevOps & Cloud: Git, Docker, Github Actions, Azure, AWS • Data Visualization: PowerBI, Matplotlib, Seaborn, MATLAB, OpenCV, Open3D	• Agile development • Ownership mindset • Learning agility • Problem solving • Coaching & mentoring • Analytical mindset • Business argument

Education

Peking University PhD in Artificial Intelligence	Beijing, China
<i>QS World University Rankings #14 School of Intelligence Science and Technology</i>	<i>Sep. 2016 – Jul. 2024</i>
Peking University BSc in Artificial Intelligence	Beijing, China
<i>School of Electronic Engineering and Computer Science</i>	<i>Sep. 2012 – Jul. 2016</i>
Kyushu University Visiting Scholar Program	Fukuoka, Japan
<i>Graduate School of Information Science and Electrical Engineering</i>	<i>Sep. 2018 – Mar. 2019</i>

Professional Experience

Machine Learning Engineer Yureca Technology	<i>Oct. 2024 – Now</i>
<i>Tech Stack: React Native, FastAPI, PostgreSQL, Github Actions, Azure, AWS, LLMs, RAG, VectorDB</i>	
• Led the development of an end-to-end household inventory management system, delivering data-driven insights through automated recommendations and reports. • Engineered a production-ready full-stack architecture across mobile front end, backend services, and PostgreSQL persistence, with JIRA-managed agile delivery. • Developed machine learning models to learn user consumption patterns and item features, enabling purchase cycle prediction, restock recommendations, and automatic item annotation, with online adaptation via user behavior tracking and CI/CD pipelines. • Implemented GenAI and RAG pipelines to support natural language item entry and proactive, context-aware recommendations based on user inventory state. • Designed ETL pipelines, dashboards, and periodic reports, integrating backend metrics to monitor system performance, resource usage, and cost efficiency.	

Software Engineer | CLCN

Mar. 2025 – Now

Tech Stack: React Vite, Node.js, Figma, Github Actions

- ♦ Contribute to **the front-end development of an internal ERP web app** that creates a transparent, measurable collaboration framework and standardizes handoffs.
- ♦ Transform manual processes into **automated, step-based workflows** that deliver guidance, progress tracking, and reminders, aligned with clear UI flows/specs defined by business and design teams.
- ♦ **Build collaboration features around volunteer profiles and availability** so organizers can assemble teams, send invitations, and coordinate tasks transparently.
- ♦ Establish **unit and component tests for critical flows**, define QA checklist and regression suites.
- ♦ Instrument **data capture with data tracking** to support post-event reviews and training, laying the foundation for analytics-driven improvements and policy updates.

Before the completion of PhD

Machine Learning Engineer | SenseTime

Sep. 2020 – Sep. 2021

- ♦ Built an automated multi-sensor data pipeline for **mobile 3D capture**, including standardizing images, time-aligning IMU data, and extracting features, to power downstream 3D reconstruction models.
- ♦ Acted as a professional consultant, engaging stakeholders for **requirement collection and success-metric alignment, setting acceptance thresholds and release gates to app performance**.
- ♦ Developed a **deep-learning multi-view model** to reduce reconstruction uncertainty, and enabled a **post-deployment fine-tuning loop** on user-returned images.
- ♦ Optimized inference efficiency by integrating classical 3D matching algorithms with deep learning models, **reducing computational load for mobile deployment**.
- ♦ **Delivered the solution as a modular Python package with clear docs and tests**, and exposed a lightweight REST/HTTP wrapper to streamline integration with the mobile app and internal tooling, enabled downstream 3D-model-based features.

Computer Vision Engineer | BOE Technology

Sep. 2018 – Sep. 2019

- ♦ Coordinated cross-functional teams to develop a **visual navigation system for autonomous cleaning robots**, aligning perception, planning and hardware modules.
- ♦ Installed and calibrated stereo cameras and supportive sensors, performing intrinsic and extrinsic calibration to **achieve sub-millimetre pose estimation accuracy in production environments**.
- ♦ Implemented **C++ modules for real-time sensor data acquisition and preprocessing**, improving data reliability under varying lighting conditions, integrated into the commercial cleaning-robot line.
- ♦ Produced technical documentation and conducted workshops to help engineering and operations teams **interpret system outputs and safely adjust parameters** for different deployment scenarios.

Research Achievements

PhD research in **robotics and 3D perception**, with a focus on modeling **uncertainty in real-world sensor data**. Designed efficient deep learning methods that balance **accuracy, robustness, and computational cost**, enabling reliable perception under practical system constraints.

Publications

- ♦ **Rukun Qiao**, Hiroshi Kawasaki and Hongbin Zha, "TIDE: Temporally Incremental Disparity Estimation via Pattern Flow in Structured Light System," in *IEEE Robotics and Automation Letters*, pp.5111-5118, April 2022.
- ♦ **Rukun Qiao**, Hiroshi Kawasaki and Hongbin Zha, "Online Adaptive Disparity Estimation for Dynamic Scenes in Structured Light Systems," in *IEEE/RSJ International Conference on Intelligent Robotics and Systems*, 2023.
- ♦ **Rukun Qiao**, Hiroshi Kawasaki and Hongbin Zha, "Depth Estimation in Structured Light Systems via Neural Implicit Functions," in *International Conference on 3D Vision*, 2024.